

**Time:** 5 minutes**Outcome:** Apply the textbook fractional-reserve deposit-creation process.**Difficulty:** ●●○ Intermediate

## Deposit Creation and the Money Multiplier



### THE CORE IDEA

In the simple textbook model, banks keep the required fraction of deposits as reserves and lend the rest. When loan proceeds are redeposited, another bank can lend part again. The repeated process creates deposits. The simple money multiplier  $m = 1 / rr$ , where  $rr$  is the required reserve ratio, gives the theoretical maximum total deposit expansion per dollar of new reserves under full lending and redepositing.



### HOW TO RECOGNIZE IT

- Required reserves = reserve ratio x deposits; excess reserves can support new lending.
- A lower reserve ratio produces a larger simple multiplier.
- Each lending round is smaller because a fraction is retained as reserves.
- Currency holding and excess reserves make actual expansion smaller than the maximum.



### WATCH OUT

Do not confuse the multiplier with bank capital or claim that one bank lends the entire multiplied amount. The multiplier summarizes repeated system-wide rounds under restrictive assumptions.



### WORKED EXAMPLE: A THEORETICAL MAXIMUM

Textbook deposit rounds: reserve ratio 10%

|                  | New deposit | Reserves | New loan |
|------------------|-------------|----------|----------|
| First bank       | \$1,000     | \$100    | \$900    |
| Second bank      | \$900       | \$90     | \$810    |
| Third bank       | \$810       | \$81     | \$729    |
| All rounds total | \$10,000    | \$1,000  | \$9,000  |

With  $rr = 10\%$ , the simple multiplier is  $1 / 0.10 = 10$ . If \$1,000 of new reserves enters the banking system, maximum total deposits are \$10,000. Banks collectively retain \$1,000 as required reserves and can support up to \$9,000 of new loans. Cash holding or excess reserves lowers the result.



### CHECK YOURSELF

If  $rr$  rises from 10% to 20%, what happens to the simple multiplier?

**READY?**

Return to the game and master this concept.